



**POLITECNICO
MILANO 1863**



Design Document

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1. Introduction

1.1 Purpose

Triplo is a cross-platform mobile application for smartphones and smartwatches, conceived as a digital logbook for trekking activities.

The application aims to enhance the experience of hiking in mountain environments by combining route information, environmental data, and recording of personal activity.

Users can explore predefined trekking routes and access useful information about the paths, including navigation support and contextual data such as weather conditions.

At the same time, the application allows hikers to document their experiences by creating logbook entries enriched with photos and personal diary notes.

The platform also enables users to discover and follow other hikers, and to browse selected entries from their trekking diaries, enhancing the sharing of experiences and inspiration for future hikes.

During a hike, users can optionally complete challenges, such as time-based goals designed to enrich the trekking experience.

The following sections describe the system architecture, the main functionalities of the application, and the technological choices adopted in its development.

1.2 Scope

Triplo supports users during trekking activities by providing guidance, contextual information, and tools to document their experiences in mountain environments.

The application focuses on the Madonna di Campiglio area and offers a set of predefined trekking routes with different levels of difficulty.

During a hike, users can access information related to the selected route, including navigation support and environmental conditions that may influence the trekking experience.

The application also encourages engagement through optional challenges that users may complete along the route. Each trekking activity can be recorded in a personal digital logbook, where users can document their experience by adding photos and diary notes. Users can also browse selected entries from the logbooks of other hikers they follow, allowing them to discover new routes and gain inspiration from shared experiences.

The application aims to support outdoor exploration while enabling users to record, revisit, and share meaningful moments from their trekking activities.

2. Architectural Design

2.1 Overview